UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Types of Electrodes

Complete study pack for class 003: the six types of electrodes (metal–metal ion, metal–insoluble salt–anion, gas, amalgam, oxidation–reduction with the quinhydrone electrode, and ion-selective membrane electrodes), each with its representation, electrode reaction, Nernst equation and an example

Compiled from the PESU class 003 slide deck (titled “Class 16 content”) and its AV-summary transcript, the older consolidated Unit 2 slide PDF, and ten deduplicated past ESA papers. Generated 17 October 2026.

· How to use this pack

Topic 1 is the map: the classification tree, the comparison of all six types and the one procedure for writing any electrode's Nernst equation. Topics 2–7 take one type each in the deck's order. Every type has the same four parts: representation, electrode (reduction) reaction, Nernst equation and an example. The numericals are short and verified with sympy; standard potentials that the deck does not give are marked as given data. Reference electrodes (calomel in detail) are class 004; the glass-electrode construction and pH numericals are in Unit 3. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·5	Deck 003, slide 5 (rendered .pptx)
T003 00:07:50	AV-summary transcript 003, segment start
PDF p.57	Older consolidated Unit 2 slide PDF (8enterprise), page 57
ESA Dec-23 2d	Past paper: ESA Dec 2023, question 2.d
ESA Jan-25 s1 2b	Past paper: ESA Jan 2025 set 1, question 2.b

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7. 7 · Ion-selective (membrane) electrode
8. Appendix A · Symbol glossary
9. Appendix B · Sources, coverage and gaps

1 Types of electrodes: the map

A cell needs two half-cells, so two electrodes; which electrode is used depends on the application. The deck lists six types: metal–metal ion, metal–metal insoluble salt–anion, gas, amalgam, oxidation–reduction (with the quinhydrone electrode as a special case) and ion-selective (membrane) electrodes. For every type you should know four things: the representation, the electrode reaction written as a reduction, the Nernst equation and one example. One procedure writes the Nernst equation for all of them.

Key terms and definitions

Electrode (half-cell). One of the two halves of a cell: a conductor in contact with an electrolyte at whose interface a potential develops. "In order to form a cell, 2 half cells or 2 electrodes are required."
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> *"In order to form a cell, 2 half cells or 2 electrodes are required. Various types of electrodes are available which are constructed based on the application."* S003·3

Formulas

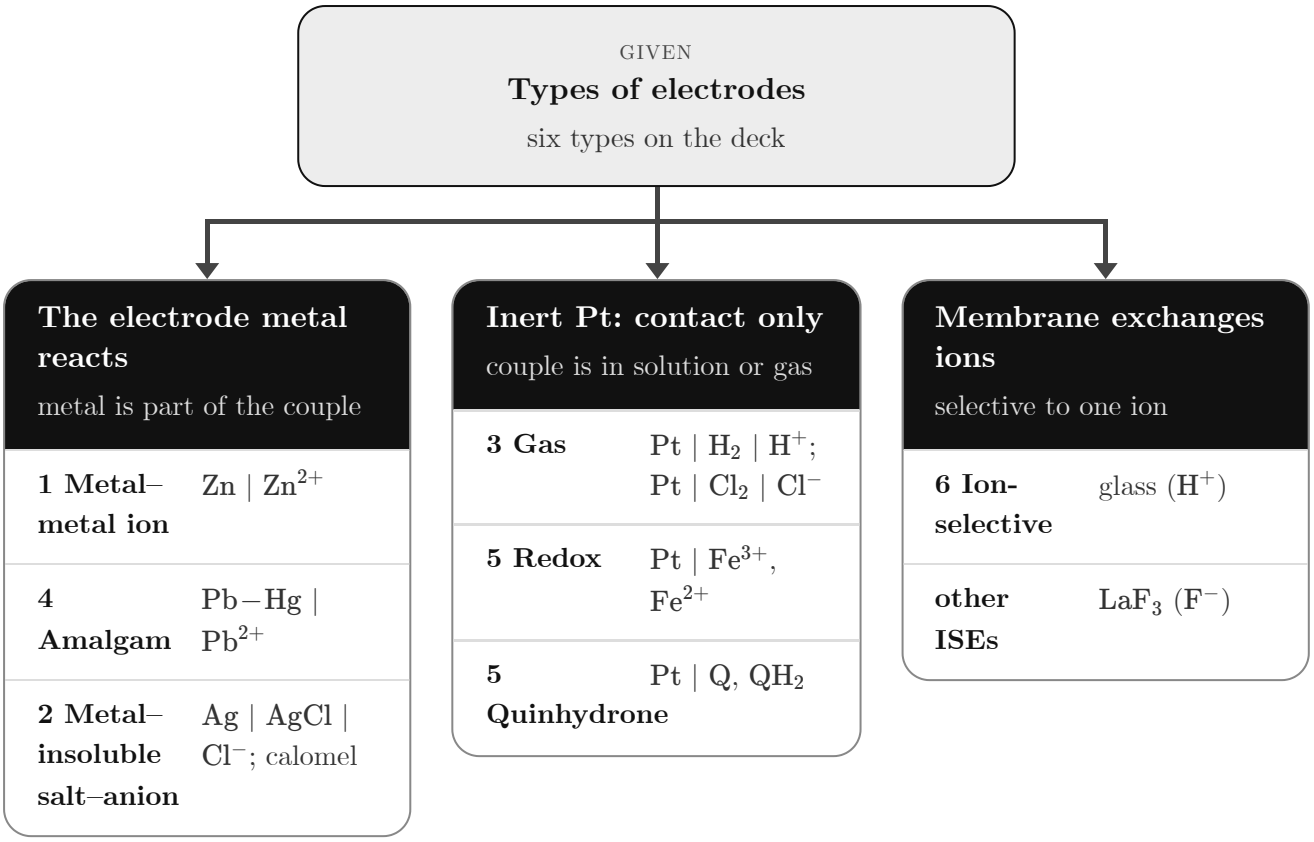
E1.1 · Nernst equation for any electrode (reduction form) ESA ★★

$$E = E^\circ - \frac{2.303RT}{nF}\log\frac{[\text{products}]}{[\text{reactants}]} = E^\circ - \frac{0.0591}{n}\log\frac{[\text{products}]}{[\text{reactants}]} \quad (298\ \text{K})$$

gives	The single-electrode (reduction) potential at any concentration.
applies	Write the electrode reaction as a reduction first; pure solids and pure metals count as 1, gases enter as partial pressure in atm.
units	V
note	$2.303RT/F = 0.0591$ V at 298 K (0.05914 with $R = 8.314$, $F = 96485$).

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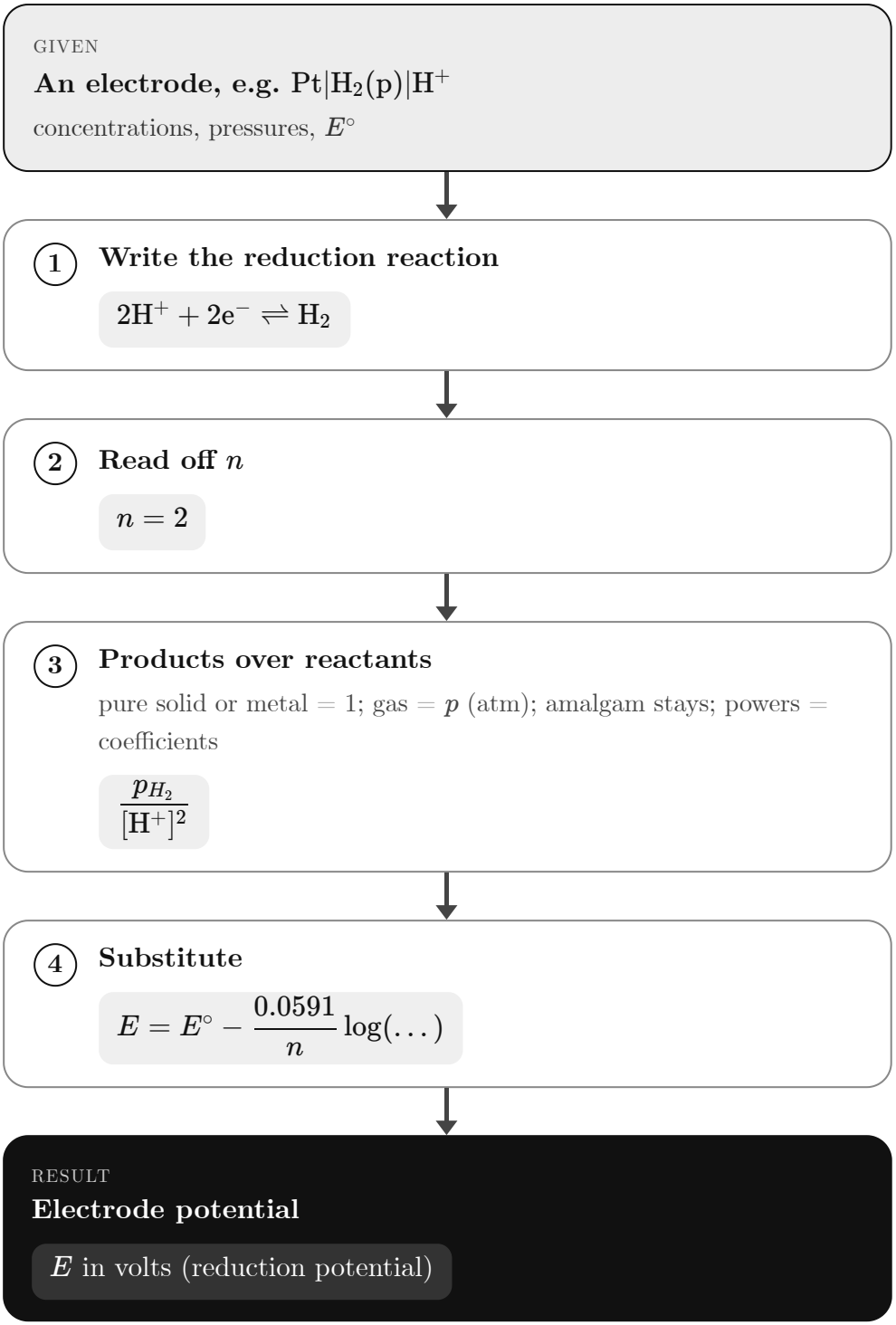
Flowcharts



Remark. The three groups are added for clarity; the numbers are the deck's. The quinhydrone electrode is a special case of type 5.

Figure F1.1. Classification of electrodes (the six types in three groups). ESA ★★

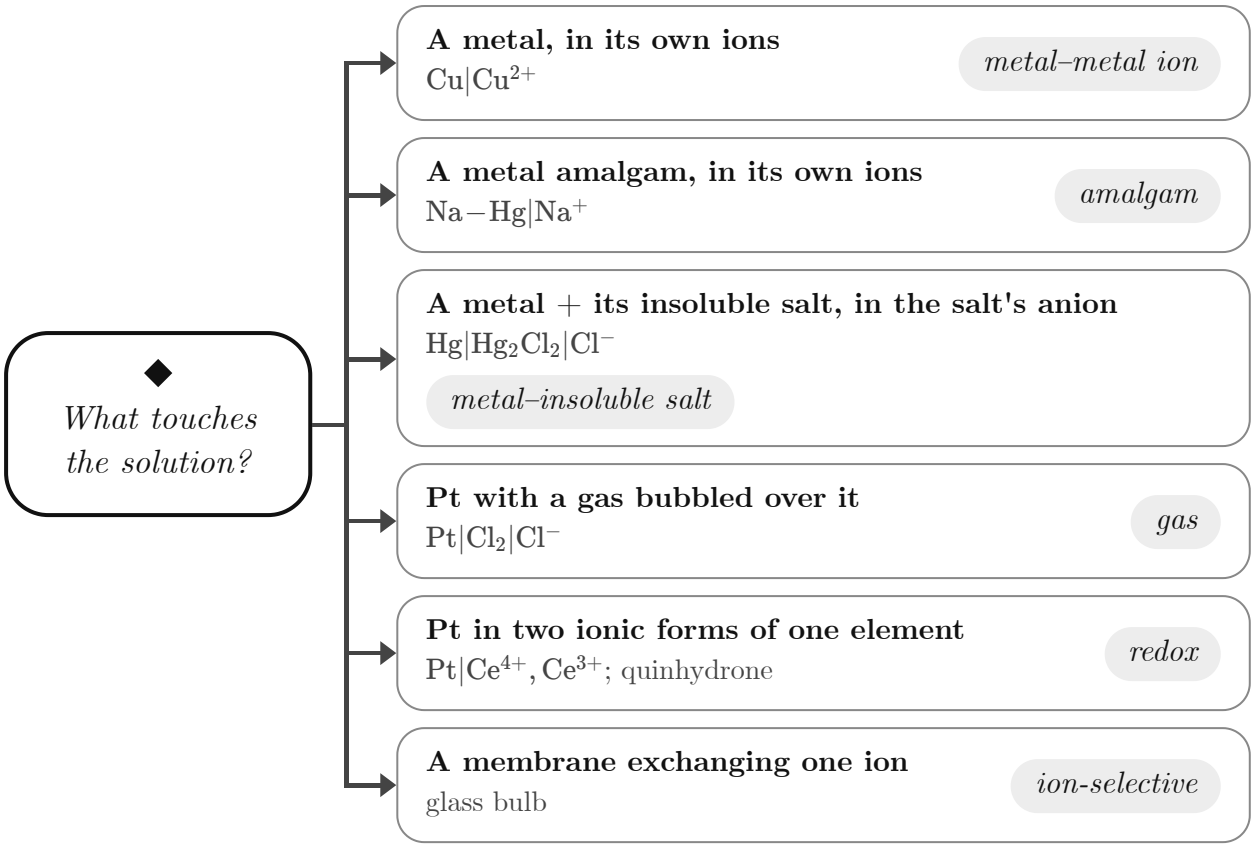
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Exam trap. Writing the oxidation instead of the reduction flips the log term and gives the wrong sign.

Figure F1.2. Writing the Nernst equation for any electrode. ESA ★★

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Remark. Pt never takes part in the reaction; if Pt appears, the electrode is gas or redox.

Figure F1.3. Which type is this electrode? (from its representation) ESA ★

S003·3 · S003·4 · S003·5 · S003·6 · S003·7 · S003·9

The six types of electrodes at a glance

Type (responds to)	Representation	Electrode reaction	Nernst equation, 298 K
1 Metal–metal ion *(cation)*	$\mathrm{Zn}\|\mathrm{Zn^{2+}}$	$\mathrm{M^{n+} + ne^- \rightleftharpoons M}$	$E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$
2 Metal–insoluble salt–anion *(anion)*	$\mathrm{Ag}\|\mathrm{AgCl(s)}\|\mathrm{Cl^-}$	$\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$	$E^\circ - 0.0591\log[\mathrm{Cl^-}]$
3 Gas *($\mathrm{H^+}$, p)*	$\mathrm{Pt}\|\mathrm{H_2}\|\mathrm{H^+}$	$\mathrm{2H^+ + 2e^- \rightleftharpoons H_2}$	$E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[\mathrm{H^+}]^2}$
4 Amalgam *(cation, amalgam)*	$\mathrm{Pb-Hg}\|\mathrm{Pb^{2+}}$	$\mathrm{Pb^{2+} + 2e^- \rightleftharpoons Pb(Hg)}$	$E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Pb\text{-}Hg}]}{[\mathrm{Pb^{2+}}]}$
5 Redox *(ratio red/ox)*	$\mathrm{Pt}\|\mathrm{Sn^{4+}, Sn^{2+}}$	$\mathrm{Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}}$	$E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$
5 Quinhydrone *(pH)*	$\mathrm{Pt}\|\mathrm{Q, QH_2}$	$\mathrm{Q + 2H^+ + 2e^- \rightleftharpoons QH_2}$	$0.699 - \frac{0.0591}{2}\log\frac{[\mathrm{QH_2}]}{[Q][\mathrm{H^+}]^2}$
6 Ion-selective (glass) *(one ion, $\mathrm{H^+}$)*	glass membrane	ion exchange at the membrane	$E^\circ_G + 0.0591\log[\mathrm{H^+}]$

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Important points and why

1. Always write the electrode reaction as a **reduction**, because electrode potentials are reduction potentials; the Nernst log term is then products over reactants. ESA ★★ T003 00:01:38 · T003 00:07:50
2. Pure solids and pure metals have activity 1 and drop out of the log term; gases enter as partial pressure (atm); amalgams stay in. ESA ★★ T003 00:11:06 · T003 00:08:07
3. 0.0591 in the equation means room temperature (298 K): $2.303RT/F$. ESA ★ T003 00:05:12
4. Three groups: the electrode metal reacts (metal–ion, amalgam, metal–insoluble salt); an inert Pt only carries electrons (gas, redox, quinhydrone); a membrane exchanges ions (ion-selective). ESA ★ ADDED FOR CLARITY S003·3 · S003·9

SHORT-NOTE RECAP

A cell needs two electrodes; the type depends on the application. Six types: metal–metal ion, metal–insoluble salt–anion, gas, amalgam, redox (quinhydrone a special case), ion-selective. For each, write the reduction reaction and $E = E^\circ - \frac{0.0591}{n}\log\frac{[\text{products}]}{[\text{reactants}]}$ (solids 1, gases as p).

2 Metal–metal ion electrode

A metal rod dips in a solution of its own ions, e.g. $\mathrm{Zn|Zn^{2+}}$, $\mathrm{Cu|Cu^{2+}}$, $\mathrm{Ag|Ag^{+}}$. A potential develops at the metal–solution interface. The reaction is $\mathrm{M^{n+} + ne^- \rightleftharpoons M}$ and the pure metal has activity 1, so the potential depends only on $[\mathrm{M^{n+}}]$: $E = E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$.

Key terms and definitions

Metal–metal ion electrode. Metal in contact with a solution of its own ions, e.g. $\mathrm{Zn/Zn^{2+}}$, $\mathrm{Cu/Cu^{2+}}$, $\mathrm{Ag/Ag^{+}}$. S003·3 · T003 00:00:59

Formulas

E2.1 · Metal–metal ion electrode ESA ★

$$E_{\mathrm{M^{n+}/M}} = E^\circ_{\mathrm{M^{n+}/M}} + \frac{2.303RT}{nF}\log[\mathrm{M^{n+}}]$$

gives	Potential of $\mathrm{M|M^{n+}}$.
applies	$\mathrm{M^{n+} + ne^- \rightleftharpoons M}$; pure metal, activity 1.
units	V
note	Same as $E^\circ - \frac{0.0591}{n}\log\frac{1}{[\mathrm{M^{n+}}]}$: the + sign comes from inverting the log.

S003·3 · T003 00:02:15

Graphs and diagrams

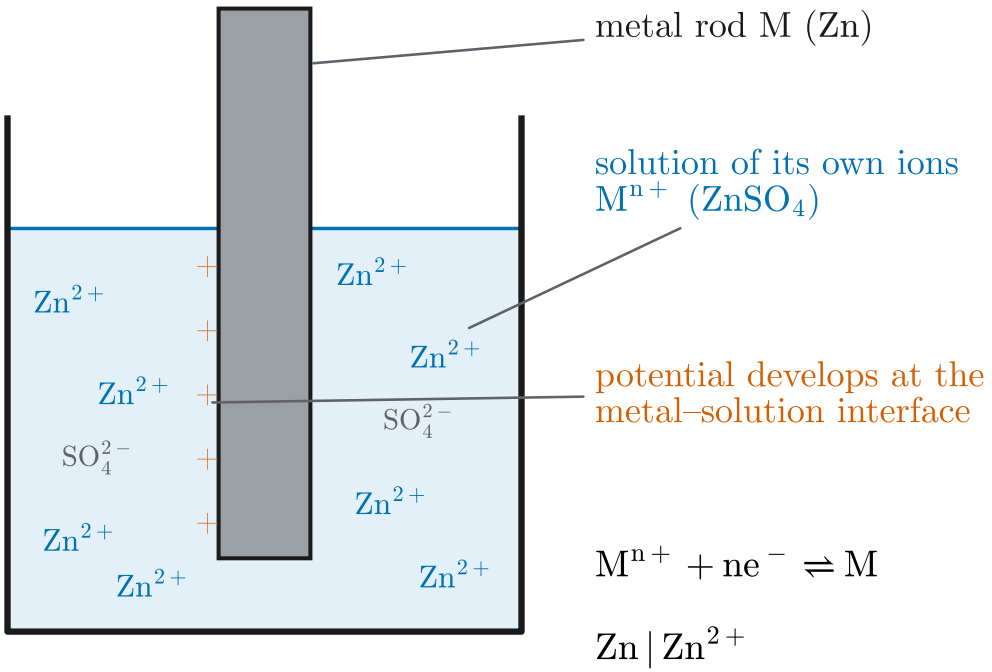


Figure G2.1. Metal–metal ion electrode ($\mathbf{Zn|Zn^{2+}}$). ESA ★

S003·3 · T003 00:01:26

How to read it. A zinc rod dips in a solution of its own ions. The potential develops at the rod–solution interface (marked +). Label: metal rod, solution of its own ions, the interface; write the reaction and the representation under it. Schematic, not to scale.

Important points and why

1. Examples: $\mathbf{Zn|Zn^{2+}}$, $\mathbf{Cu|Cu^{2+}}$, $\mathbf{Ag|Ag^{+}}$; reaction $M^{n+} + ne^- \rightleftharpoons M$. ESA ★ S003·3 · T003 00:01:13
2. Increasing $[M^{n+}]$ tenfold raises E by $0.0591/n$ V. ESA ★ ADDED FOR CLARITY S003·3

Why is there a + sign in $E = E^\circ + \frac{0.0591}{n}\log[M^{n+}]$, when the others have −?
For $M^{n+} + ne^- \rightleftharpoons M$ the log term is $\log \frac{1}{[M^{n+}]} = -\log[M^{n+}]$; the two minus signs make +. Every electrode follows the same rule E1.1.

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Worked examples

W2.1 · Zinc electrode in 0.01 M Zn^{2+} ESA ★

GIVEN $Zn|Zn^{2+}(0.01\,M)$, $E^\circ_{Zn^{2+}/Zn} = -0.76$ V (given), 298 K.

STEP 1 $Zn^{2+} + 2e^- \rightleftharpoons Zn$, $n = 2$.

STEP 2 $E = E^\circ + \frac{0.0591}{2}\log[Zn^{2+}] = -0.76 + 0.02955\log 0.01 = -0.76 + 0.02955(-2)$.

ANSWER $\boxed{E = -0.8191\ \text{V}}$

CHECK sympy: −0.819100 V. Dilution makes the electrode more negative.

ADDED FOR CLARITY S003·3

SHORT-NOTE RECAP

Metal in contact with a solution of its own ions ($Zn|Zn^{2+}$, $Cu|Cu^{2+}$, $Ag|Ag^+$). $M^{n+} + ne^- \rightleftharpoons M$; $E = E^\circ + \frac{2.303RT}{nF}\log[M^{n+}]$.

3 Metal–metal insoluble salt–anion electrode

A metal is in contact with a sparingly soluble salt of the same metal, dipped in a solution of a soluble salt of the same anion. Examples: the calomel electrode $\mathrm{Hg|Hg_2Cl_2|KCl}$ and the silver–silver chloride electrode $\mathrm{Ag|AgCl(s)|HCl}$. For $\mathrm{Ag/AgCl}$, $\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$ and $E = E^\circ - 0.0591 \log[\mathrm{Cl^-}]$: the electrode responds to the anion, not to the metal ion.

Key terms and definitions

Metal–metal insoluble salt–anion electrode. These electrodes consist of a metal in contact with a sparingly soluble salt of the same metal dipped in a solution of soluble salt of the same anion, e.g. calomel electrode $\mathrm{Hg/Hg_2Cl_2/KCl}$, $\mathrm{Ag/AgCl(s)/HCl}$. S003·4 · T003 00:02:31

Formulas

E3.1 · Silver–silver chloride electrode ESA ★

$$E_{\mathrm{Ag/AgCl/Cl^-}} = E^\circ_{\mathrm{Ag/AgCl/Cl^-}} - \frac{0.0591}{1} \log[\mathrm{Cl^-}]$$

gives	Potential of $\mathrm{Ag|AgCl(s)|Cl^-}$.
applies	$\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$; $n = 1$.
units	V
note	Reversible to $\mathrm{Cl^-}$: E falls as $[\mathrm{Cl^-}]$ rises. $E^\circ \approx 0.222$ V (given data).

S003·4 · T003 00:04:57

E3.2 · Calomel electrode (same type) ESA ★

$$\mathrm{Hg_2Cl_2 + 2e^- \rightleftharpoons 2Hg + 2Cl^-}, \qquad E = E^\circ - \frac{0.0591}{2} \log[\mathrm{Cl^-}]^2 = E^\circ - 0.0591 \log[\mathrm{Cl^-}]$$

gives	Potential of $\mathrm{Hg|Hg_2Cl_2(s)|KCl}$.
applies	Reference electrode of class 004.
units	V
note	SCE (saturated KCl) 0.2422 V, 1 M 0.2824 V, 0.1 M 0.3358 V at 298 K (consolidated PDF).

not on current slides S003·4 · PDF p.33

Graphs and diagrams

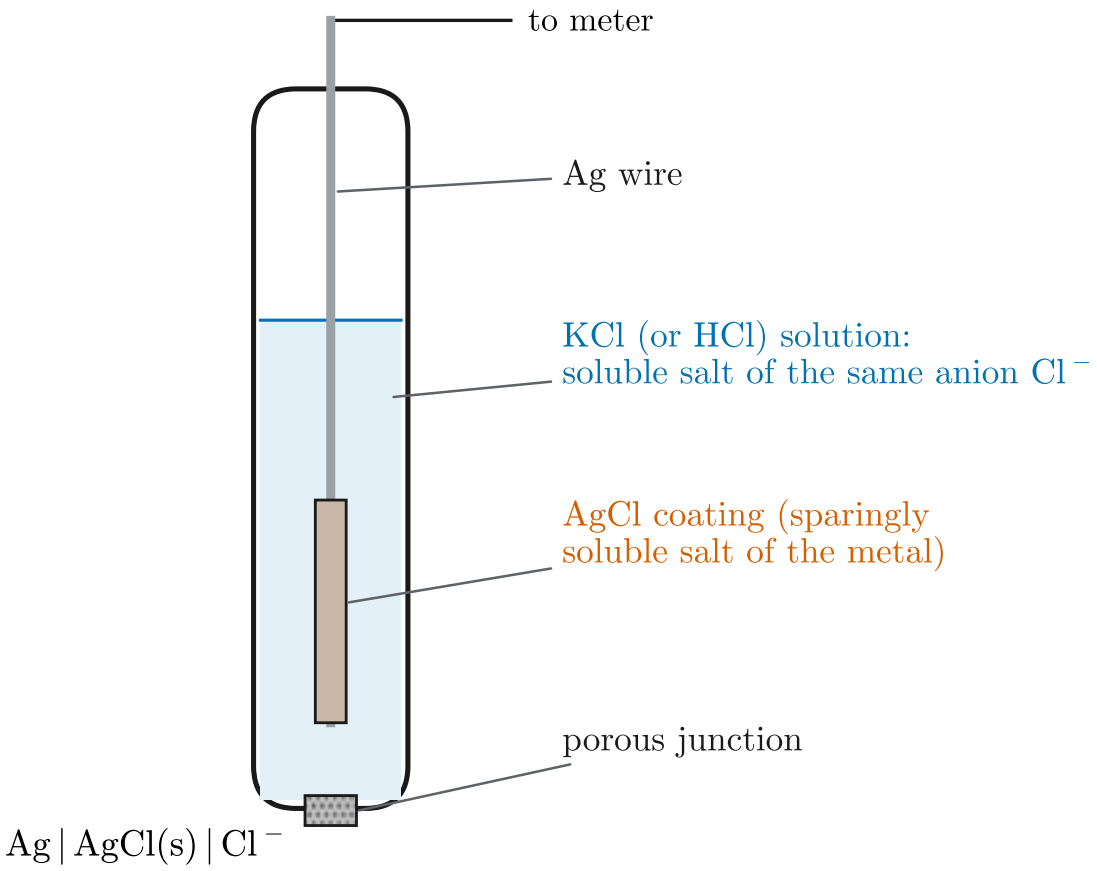


Figure G3.1. Silver–silver chloride electrode. ESA ★

S003·4 · T003 00:04:34

> ***How to read it.*** A silver wire, its lower end coated with AgCl, sits in KCl (or HCl) inside a glass tube that meets the outer solution through a porous junction. The three labels to remember: metal, sparingly soluble salt of the metal, soluble salt of the same anion.

Important points and why

1. The electrode reaction product is **Ag**, not Ag^+: $AgCl + e^- \rightleftharpoons Ag + Cl^-$. With Ag^+ the equation would not even balance. ESA ★★ T003 00:04:20
2. The solution must contain the anion of the sparingly soluble salt (Cl^- from KCl or HCl), so the potential depends on $[Cl^-]$: these electrodes are reversible to the anion. ESA ★★ T003 00:04:50 · PDF p.33
3. Construction of **Ag/AgCl**: a silver wire coated with AgCl dipped in KCl (or HCl), with a porous junction. ESA ★ S003·4 · T003 00:04:34
4. Both common reference electrodes (calomel and **Ag/AgCl**) are of this type; class 004 covers them. ESA ★ S003·4 · T003 00:18:34

Exam trap. Writing $\mathrm{AgCl + e^- \rightleftharpoons Ag^+ + Cl^-}$ is wrong (and unbalanced in charge); the product is metallic Ag. T003 00:04:20

Exam trap. The Ag/AgCl and calomel potentials fall as $[\mathrm{Cl^-}]$ rises (minus sign); students often write $+$. ADDED FOR CLARITY S003·4

Worked examples

W3.1 · Ag/AgCl electrode at $[\mathrm{Cl^-}] = 0.1$ M

ESA ★★

GIVEN $\mathrm{Ag|AgCl(s)|Cl^-(0.1\,M)}$, $E^\circ_{\mathrm{Ag/AgCl/Cl^-}} = 0.222$ V (given), 298 K.

STEP 1 $\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$, $n = 1$.

STEP 2 $E = 0.222 - 0.0591 \log[\mathrm{Cl^-}] = 0.222 - 0.0591 \log 0.1 = 0.222 + 0.0591.$

ANSWER $\boxed{E = 0.2811\ \mathrm{V}}$

CHECK sympy: 0.281100 V. Lower $[\mathrm{Cl^-}]$ than 1 M, so $E > E^\circ$.

S003·4 · PDF p.67

SHORT-NOTE RECAP

Metal + sparingly soluble salt of the metal, in a solution of a soluble salt of the same anion. Calomel $\mathrm{Hg|Hg_2Cl_2|KCl}$; $\mathrm{Ag|AgCl(s)|HCl}$: $\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$, $E = E^\circ - 0.0591 \log[\mathrm{Cl^-}]$. Reversible to the anion; used as reference electrodes.

4 Gas electrode

A gas bubbles around an inert metal foil (platinum) dipped in a solution of the ions to which the gas is reversible. The platinum provides electrical contact and helps set up the gas–ion equilibrium. Examples: the hydrogen electrode $\mathrm{Pt}|\mathrm{H_2}|\mathrm{H^+}$ and the chlorine electrode $\mathrm{Pt}|\mathrm{Cl_2}|\mathrm{Cl^-}$. In the Nernst equation the gas enters as its partial pressure: $E = E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[H^+]^2}$. Asked in ESA Dec 2023 as a 2-mark note.

Key terms and definitions

Gas electrode. It consists of gas bubbling about an inert metal foil, immersed in solution containing ions to which the gas is reversible. The metal provides electrical contact and facilitates the establishment of equilibrium between the gas and its ions, e.g. hydrogen electrode $\mathrm{Pt}/\mathrm{H_2}/\mathrm{H^+}$, chlorine electrode $\mathrm{Pt}/\mathrm{Cl_2}/\mathrm{Cl^-}$. S003·5 · T003 00:05:23

> *"The metal provides electrical contact and facilitates the establishment of equilibrium between the gas and its ions."* S003·5

Formulas

E4.1 · Hydrogen electrode ESA ★★

$$E_{\mathrm{Pt}/\mathrm{H_2}/\mathrm{H^+}} = E^\circ_{\mathrm{Pt}/\mathrm{H_2}/\mathrm{H^+}} - \frac{0.0591}{2}\log\frac{p_{\mathrm{H_2}}}{[\mathrm{H^+}]^2}$$

gives	Potential of $\mathrm{Pt}\|\mathrm{H_2(p)}\|\mathrm{H^+}$.
applies	$2\mathrm{H^+} + 2\mathrm{e^-} \rightleftharpoons \mathrm{H_2}$; $n = 2$; $E^\circ = 0$ by convention.
units	V
note	SHE: $p_{H_2} = 1$ atm, $[\mathrm{H^+}] = 1$ M, so the log term is 0 and $E = E^\circ = 0$.

S003·5 · T003 00:07:50 · T003 00:08:38

E4.2 · Hydrogen electrode in terms of pH ESA ★★

$$E = -\frac{0.0591}{2}\log p_{\mathrm{H_2}} - 0.0591\,\mathrm{pH} \qquad (p_{\mathrm{H_2}} = 1\ \mathrm{atm}:\ E = -0.0591\,\mathrm{pH})$$

gives	The pH dependence of the hydrogen electrode.
applies	From E4.1 with $E^\circ = 0$ and $\mathrm{pH} = -\log[\mathrm{H^+}]$.
units	V
note	A hydrogen electrode can therefore measure pH.

ADDED FOR CLARITY S003·5

E4.3 · Chlorine electrode ESA ★

$$Cl_2 + 2e^- \rightleftharpoons 2Cl^-, \qquad E = E^\circ - \frac{0.0591}{2} \log \frac{[Cl^-]^2}{p_{Cl_2}}$$

GIVES	Potential of $Pt|Cl_2(p)|Cl^-$.
APPLIES	$n = 2$; $E^\circ = 1.36$ V (given data).
UNITS	V
NOTE	The slide names the chlorine electrode but gives only the hydrogen equation.

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Graphs and diagrams

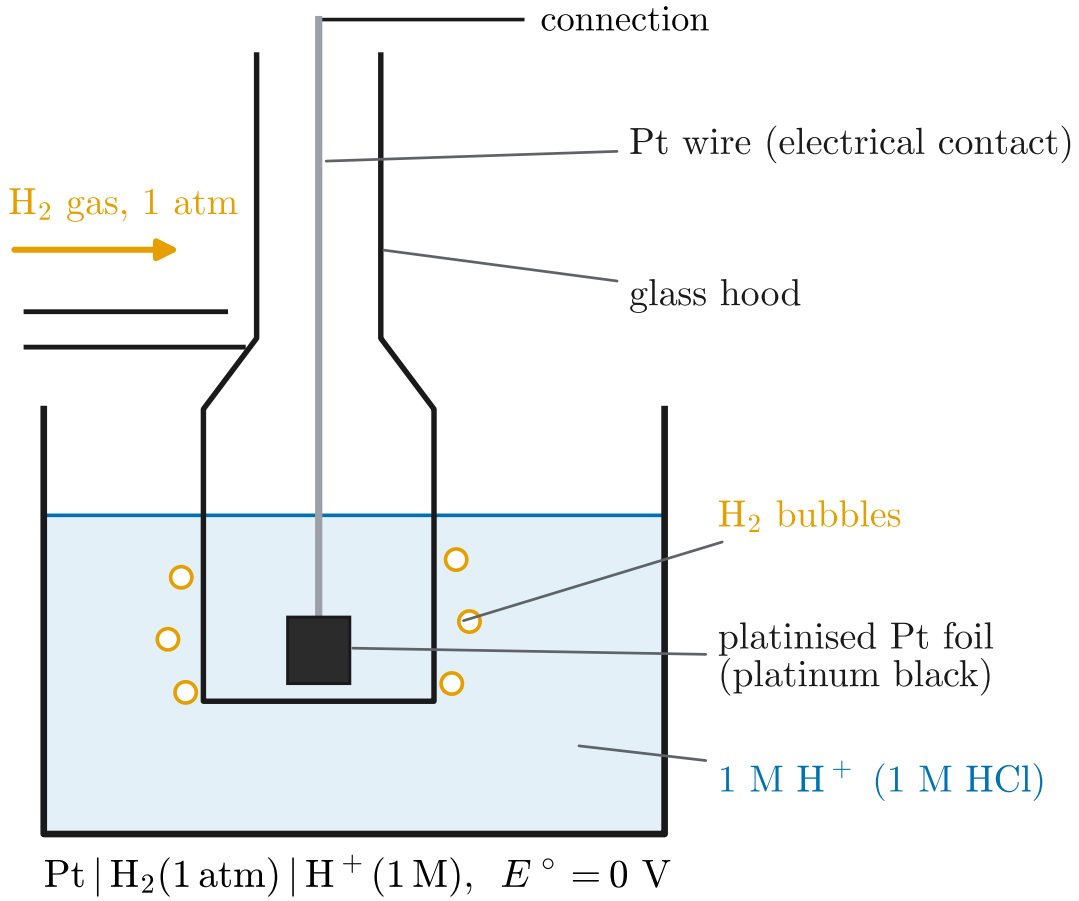


Figure G4.1. Standard hydrogen electrode (a gas electrode). ESA ★★

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How to read it. Hydrogen at 1 atm enters a glass hood and bubbles over a platinised Pt foil dipped in 1 M HCl; the Pt wire leads out to the connection. Label: H_2 inlet, Pt wire, platinised foil, 1 M H^+, bubbles. Redrawn from the deck's figure; schematic.

Important points and why

1. Pt is chosen because it is inert; platinised (platinum black) foil gives a large surface for the gas–ion equilibrium. ESA ★★ S003·5 · T003 00:06:07
2. The gas must be bubbled into a solution of the ions to which it is reversible: $\mathrm{H_2}$ into $\mathrm{H^+}$, $\mathrm{Cl_2}$ into $\mathrm{Cl^-}$. ESA ★★ T003 00:06:45 · T003 00:07:03
3. SHE: $\mathrm{H_2}$ at 1 atm, $[\mathrm{H^+}] = 1$ M (1 M HCl), so $E = E^\circ = 0$ V; it is the primary reference. ESA ★★ S003·5 · T003 00:08:38
4. Hydrogen electrode: $E = -0.0591\,\mathrm{pH}$ at 1 atm; it becomes more negative by 0.0591 V per pH unit. ESA ★★ ADDED FOR CLARITY S003·5

Why use a partial pressure for hydrogen?
$\mathrm{H_2}$ is a gas; the activity of a gas is its partial pressure in atm. So the log term is $p_{H_2}/[\mathrm{H^+}]^2$.
T003 00:08:07 · T003 00:08:20

Why is the metal in a gas electrode platinum?
It must not react; it only conducts electrons and catalyses the gas–ion equilibrium on its (platinised) surface.
S003·5 · T003 00:06:25

Exam trap. Forgetting the square: the log term is $p_{H_2}/[\mathrm{H^+}]^2$, not $p_{H_2}/[\mathrm{H^+}]$; with $n = 2$ the pH term becomes $0.0591\,\mathrm{pH}$, not $0.0591\,\mathrm{pH}/2$. T003 00:08:20

Worked examples

W4.1 · Hydrogen electrode at pH 3 and 2 atm ESA ★★

GIVEN $\mathrm{Pt|H_2(2\,atm)|H^+}$, pH = 3 so $[\mathrm{H^+}] = 10^{-3}$ M; $E^\circ = 0$.

STEP 1 $2\mathrm{H^+} + 2\mathrm{e^-} \rightleftharpoons \mathrm{H_2}$, $n = 2$.

STEP 2 $E = 0 - \frac{0.0591}{2}\log\frac{p_{H_2}}{[\mathrm{H^+}]^2} = -0.02955\log\frac{2}{10^{-6}}$.

STEP 3 $\log(2 \times 10^6) = 6.3010$, so $E = -0.02955 \times 6.3010$.

STEP 4 Split: pH part $-0.0591 \times 3 = -0.1773$ V; pressure part $-0.02955\log 2 = -0.0089$ V.

ANSWER $\boxed{E = -0.1862\ \mathrm{V}}$

CHECK sympy: −0.186195 V; both routes agree (E4.2).

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W4.2 · Chlorine electrode ESA ★

GIVEN $\mathrm{Pt|Cl_2(1\,atm)|Cl^-(0.01\,M)}$, $E^\circ = 1.36$ V (given).

STEP 1 $\mathrm{Cl_2 + 2e^- \rightleftharpoons 2Cl^-}$, $n = 2$.

STEP 2 $E = 1.36 - \frac{0.0591}{2}\log\frac{(0.01)^2}{1} = 1.36 - 0.02955(-4) = 1.36 + 0.1182$.

ANSWER $\boxed{E = 1.4782\ \text{V}}$

CHECK sympy: 1.47820 V.

ADDED FOR CLARITY S003·5

SHORT-NOTE RECAP

Gas bubbled over inert Pt foil in a solution of the ions to which the gas is reversible; Pt gives contact and helps set up the gas–ion equilibrium. $\mathrm{Pt|H_2|H^+}$: $\mathrm{2H^+ + 2e^- \rightleftharpoons H_2}$, $E = E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[H^+]^2}$; SHE (1 atm, 1 M) $E = 0$. Also $\mathrm{Pt|Cl_2|Cl^-}$.

5 Amalgam electrode

The same as a metal–metal ion electrode, but the metal is an amalgam (an alloy with mercury), e.g. the lead amalgam electrode $\mathrm{Pb-Hg|Pb^{2+}}$. Amalgamation lowers the activity of a reactive metal such as sodium so that a potential can be measured, and the activity can be set by the mercury content. Because the metal is no longer pure, its concentration appears in the Nernst equation.

Key terms and definitions

Amalgam electrode. It is similar to the metal–metal ion electrode, in which a metal amalgam is in contact with a solution containing its own ions, e.g. lead amalgam electrode $\mathrm{Pb-Hg/Pb^{2+}}$. S003·6 · T003 00:08:59

Amalgam. An alloy of a metal with mercury. (transcript only) T003 00:08:59

Formulas

E5.1 · Lead amalgam electrode ESA ★

$$E_{\mathrm{Pb^{2+}/Pb\text{-}Hg}} = E^{\circ}_{\mathrm{Pb^{2+}/Pb\text{-}Hg}} - \frac{0.0591}{2}\log\frac{[\mathrm{Pb\text{-}Hg}]}{[\mathrm{Pb^{2+}}]}$$

gives	Potential of $\mathrm{Pb-Hg|Pb^{2+}}$.
applies	$\mathrm{Pb^{2+}} + 2\mathrm{e^-} \rightleftharpoons \mathrm{Pb(Hg)}$; $n = 2$.
units	V
note	Unlike a pure metal, the amalgam's activity is not 1, so it stays in the log term.

S003·6 · T003 00:10:54 · T003 00:11:06

Graphs and diagrams

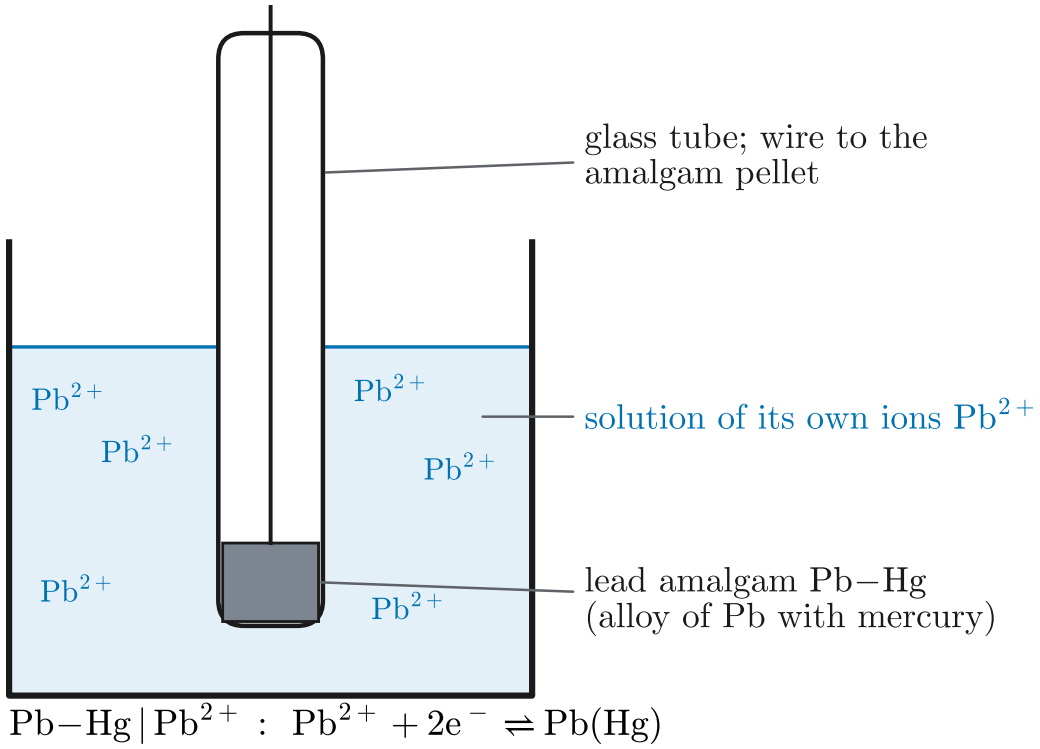


Figure G5.1. Lead amalgam electrode. ESA ★

S003·6

How to read it. A pellet of lead amalgam at the end of a glass tube (with a contact wire) dips in a solution of Pb^{2+}. It is a metal–metal ion electrode with the metal replaced by its amalgam. Simplified from the deck's figure.

Important points and why

1. Why amalgamate? Sodium in Na^{+} solution reacts vigorously, so no potential can be measured; sodium amalgam in Na^{+} works, and the mercury content controls the activity. ESA ★
   TRANSCRIPT ONLY T003 00:10:14 · T003 00:10:36
2. The amalgam's activity depends on the percentage of lead, so $[Pb-Hg]$ is kept in the log term. ESA ★ T003 00:11:24 · S003·6

Worked examples

W5.1 · Lead amalgam electrode ESA ★

GIVEN $\mathrm{Pb(Hg)}|\mathrm{Pb^{2+}}(0.01\,\mathrm{M})$ with lead activity in the amalgam 0.5; take $E^\circ_{\mathrm{Pb^{2+}/Pb\text{-}Hg}} = -0.120$ V (given, illustrative).

STEP 1 $\mathrm{Pb^{2+}} + 2\mathrm{e^-} \rightleftharpoons \mathrm{Pb(Hg)}$, $n = 2$.

STEP 2 $E = E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Pb\text{-}Hg}]}{[\mathrm{Pb^{2+}}]} = -0.120 - 0.02955\log\frac{0.5}{0.01}$.

STEP 3 $\log 50 = 1.6990$, so the correction is -0.0502 V.

ANSWER $\boxed{E = -0.1702\ \mathrm{V}}$

CHECK sympy: −0.170205 V. With activity 1 (pure metal) the amalgam term would vanish.

ADDED FOR CLARITY S003·6

SHORT-NOTE RECAP

Like the metal–metal ion electrode but with the metal as an amalgam: $\mathrm{Pb(Hg)}|\mathrm{Pb^{2+}}$, $E = E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Pb\text{-}Hg}]}{[\mathrm{Pb^{2+}}]}$. Amalgamation tames reactive metals (Na) and lets the activity be controlled.

6 Oxidation–reduction electrodes and the quinhydrone electrode

An inert metal (platinum) dips in a solution containing both the oxidised and the reduced form of a redox system, e.g. $\mathrm{Pt|Fe^{3+}, Fe^{2+}}$, $\mathrm{Pt|Ce^{4+}, Ce^{3+}}$, $\mathrm{Pt|Sn^{4+}, Sn^{2+}}$. The metal is only an electrical contact; the potential comes from the tendency of one form to change into the other and depends on the ratio [reduced]/[oxidised]. The quinhydrone electrode $\mathrm{Pt|Q, QH_2}$ is a special case whose potential also depends on $[\mathrm{H^+}]$: $E^\circ = +0.699$ V. Asked in ESA Dec 2023.

Key terms and definitions

Oxidation–reduction (redox) electrode. It consists of an inert metal such as platinum immersed in a solution containing an appropriate oxidised and reduced form of a redox system. The metal merely acts as electrical contact. The potential arises due to the tendency of one form to change into the other form, e.g. $\mathrm{Pt/Fe^{2+}, Fe^{3+}}$, $\mathrm{Pt/Ce^{3+}, Ce^{4+}}$, $\mathrm{Pt/Sn^{2+}, Sn^{4+}}$. S003·7 · T003 00:11:52

Quinhydrone electrode. An inert metal such as platinum immersed in a solution containing quinone and hydroquinone; the metal merely acts as electrical contact and the potential arises due to the tendency of quinone to change to hydroquinone. Representation $\mathrm{Pt/Q, QH_2}$. S003·8 · T003 00:14:02

Quinhydrone. An equimolar (1 : 1) compound of quinone and hydroquinone; dissolving it gives $[Q] = [\mathrm{QH_2}]$, which is why the electrode's potential then depends only on pH. (added for clarity) S003·8

> *"The metal merely acts as electrical contact. The potential arises due to the tendency of one form to change in to other form."* S003·7

Formulas

E6.1 · Stannic–stannous redox electrode ESA ★★

$$E_{\mathrm{Pt/Sn^{4+}/Sn^{2+}}} = E^\circ_{\mathrm{Pt/Sn^{4+}/Sn^{2+}}} - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$$

gives	Potential of $\mathrm{Pt|Sn^{4+}, Sn^{2+}}$.
applies	$\mathrm{Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}}$; $n = 2$.
units	V
note	Only the ratio [reduced]/[oxidised] matters. $E^\circ = 0.15$ V (given data).

S003·7 · T003 00:13:31

E6.2 · Ferric–ferrous redox electrode

ESA ★★

$$\mathrm{Fe^{3+} + e^- \rightleftharpoons Fe^{2+}}, \qquad E = E^\circ - 0.0591 \log \frac{[\mathrm{Fe^{2+}}]}{[\mathrm{Fe^{3+}}]}$$

GIVES Potential of $\mathrm{Pt|Fe^{3+}, Fe^{2+}}$.

APPLIES $n = 1$; $E^\circ = 0.77$ V (given data).

UNITS V

NOTE The Pt indicator electrode of the potentiometric titration of ferrous iron (lab).

ADDED FOR CLARITY S003·7 · T003 00:13:05 · PDF p.67

E6.3 · Quinhydrone electrode

ESA ★★

$$E_{\mathrm{Pt/Q/QH_2}} = E^\circ_{\mathrm{Pt/Q/QH_2}} - \frac{0.0591}{2} \log \frac{[\mathrm{QH_2}]}{[\mathrm{Q}][\mathrm{H^+}]^2}, \qquad E^\circ_{\mathrm{Pt/Q/QH_2}} = +0.699 \text{ V}$$

GIVES Potential of $\mathrm{Pt|Q, QH_2}$.

APPLIES $\mathrm{Q + 2H^+ + 2e^- \rightleftharpoons QH_2}$; $n = 2$.

UNITS V

NOTE $\mathrm{H^+}$ is a reactant, so it sits in the denominator, squared.

S003·8 · T003 00:15:20

E6.4 · Quinhydrone electrode in terms of pH

ESA ★★

$$[\mathrm{Q}] = [\mathrm{QH_2}] \implies E = 0.699 + 0.0591 \log[\mathrm{H^+}] = 0.699 - 0.0591\,\mathrm{pH}$$

GIVES The pH dependence used to measure pH.

APPLIES Quinhydrone dissolved (1 : 1); pH below about 8.

UNITS V

NOTE $\frac{0.0591}{2} \log[\mathrm{H^+}]^2 = 0.0591 \log[\mathrm{H^+}]$.

ADDED FOR CLARITY S003·8

Graphs and diagrams

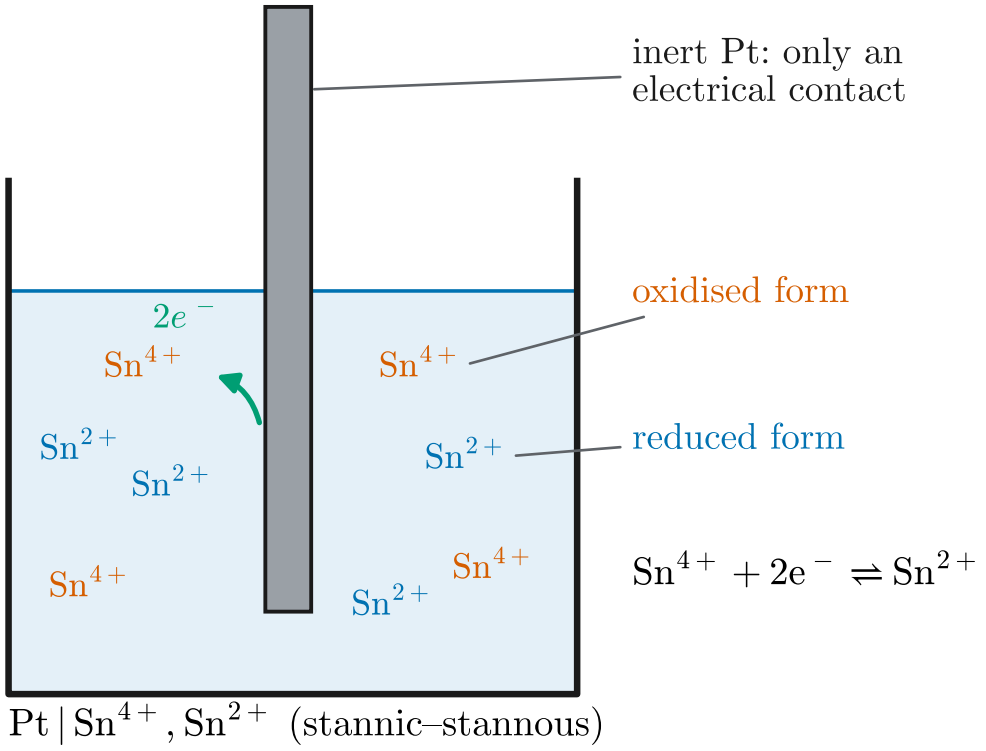


Figure G6.1. Redox electrode $\mathbf{Pt|Sn^{4+}, Sn^{2+}}$. ESA ★★

S003·7 · T003 00:12:19

How to read it. An inert Pt rod dips in a solution containing both $\mathbf{Sn^{4+}}$ (oxidised, vermilion) and $\mathbf{Sn^{2+}}$ (reduced, blue). Electrons exchanged between the two forms pass through the Pt (green arrow); the Pt itself does not react.

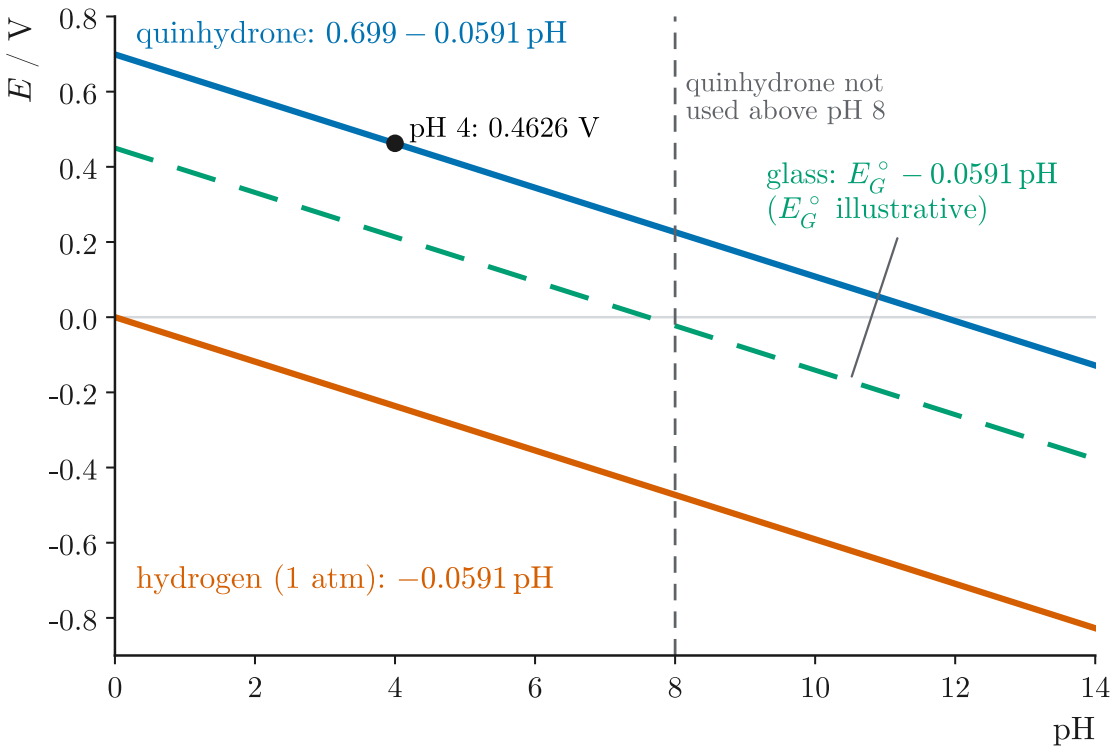
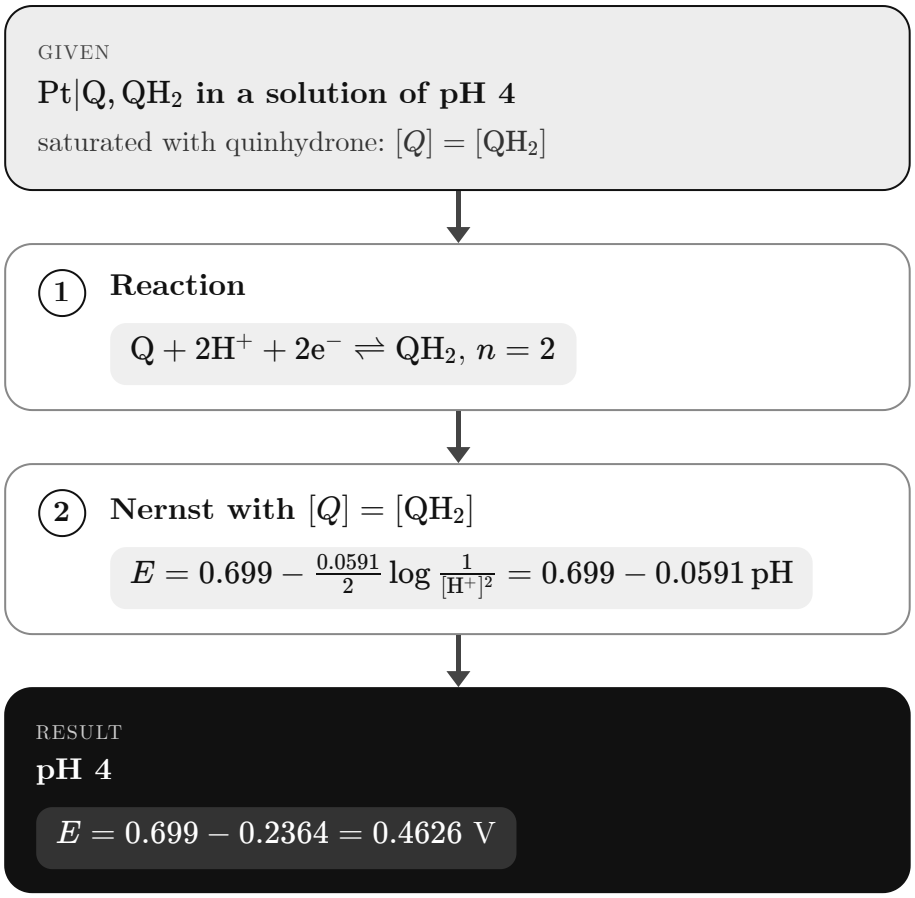


Figure G6.2. Potential against pH: hydrogen, quinhydrone and glass electrodes. ESA ★★

ADDED FOR CLARITY S003·5 · S003·8 · S003·9

How to read it. All three lines have slope −0.0591 V per pH unit at 298 K; they differ only in the intercept: 0 V (hydrogen, 1 atm), 0.699 V (quinhydrone), and E_G° (glass, illustrative value here). The dot is W6.3. The grey line marks the pH-8 limit of the quinhydrone electrode.

Flowcharts



Remark. Run it backwards to measure pH: $\mathrm{pH} = (0.699 - E)/0.0591$ (valid below about pH 8).

Figure F6.1. Potential of a quinhydrone electrode from pH. ESA ★★

S003·8

Gas electrode vs redox electrode (ESA Dec 2023 note)

	Gas electrode	Oxidation–reduction electrode
Construction	Gas bubbled over inert Pt foil in a solution of its ions	Inert Pt in a solution of both oxidised and reduced forms
Role of Pt	Contact; helps set up gas–ion equilibrium	Contact only
Example	$\mathrm{Pt}\vert\mathrm{H_2}\vert\mathrm{H^+}$, $\mathrm{Pt}\vert\mathrm{Cl_2}\vert\mathrm{Cl^-}$	$\mathrm{Pt}\vert\mathrm{Fe^{3+}},\mathrm{Fe^{2+}}$, $\mathrm{Pt}\vert\mathrm{Sn^{4+}},\mathrm{Sn^{2+}}$
Reaction	$\mathrm{2H^+ + 2e^- \rightleftharpoons H_2}$	$\mathrm{Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}}$
Nernst	$E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[\mathrm{H^+}]^2}$	$E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$
Potential set by	$[\mathrm{H^+}]$ and gas pressure	ratio [reduced]/[oxidised]

S003·5 · S003·7 · ESA Dec-23 2d

Important points and why

1. Both oxidised and reduced forms must be present in the same solution; Pt picks up the electrons exchanged between them. ESA ★★ T003 00:12:19 · T003 00:12:36
2. Unlike the gas electrode, no species is a gas and nothing is bubbled: both forms are ions in solution. ESA ★★ ADDED FOR CLARITY S003·5 · S003·7
3. Quinhydrone: $\mathrm{Q + 2H^+ + 2e^- \rightleftharpoons QH_2}$, $E^\circ = +0.699$ V; with $[Q] = [QH_2]$, $E = 0.699 - 0.0591\,\mathrm{pH}$, so it measures pH. ESA ★★ S003·8 · T003 00:14:35
4. The quinhydrone electrode is not used above about pH 8, where hydroquinone (a weak acid) ionises and is oxidised by air. ESA ★ ADDED FOR CLARITY S003·8

Why does the quinhydrone potential depend on pH, while $\mathrm{Pt}\vert\mathrm{Sn^{4+}},\mathrm{Sn^{2+}}$ does not?
$\mathrm{H^+}$ takes part in the quinone reaction ($\mathrm{Q + 2H^+ + 2e^- \rightleftharpoons QH_2}$), so it appears in the Nernst equation; the tin reaction involves no $\mathrm{H^+}$.
S003·8 · T003 00:14:35

Exam trap. In the redox Nernst equation the reduced form goes on top: $\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$. Inverting it flips the sign of the correction. S003·7 · T003 00:13:49

Exam trap. The gas and redox electrodes both use inert Pt; in a 2-mark note on either, say what Pt does (contact only) and what sets the potential (gas–ion equilibrium vs the ratio of the two ionic forms). ADDED FOR CLARITY ESA Dec-23 2d

Worked examples

W6.1 · $\mathrm{Pt|Sn^{4+}, Sn^{2+}}$ with a given ratio

ESA ★★

given $[\mathrm{Sn^{4+}}] = 0.1$ M, $[\mathrm{Sn^{2+}}] = 0.01$ M, so $[\mathrm{Sn^{2+}}]/[\mathrm{Sn^{4+}}] = 0.1$; $E^\circ = 0.15$ V (given).

step 1 $\mathrm{Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}}$, $n = 2$.

step 2 $E = 0.15 - \frac{0.0591}{2} \log \frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]} = 0.15 - 0.02955 \log 0.1 = 0.15 + 0.02955.$

answer $\boxed{E = 0.1796 \text{ V}}$

check sympy: 0.179550 V. If the ratio were reversed (10), $E = 0.1205$ V.

S003·7 · T003 00:13:49

W6.2 · $\mathrm{Pt|Fe^{3+}, Fe^{2+}}$ electrode

ESA ★★

given $[\mathrm{Fe^{3+}}] = 0.1$ M, $[\mathrm{Fe^{2+}}] = 0.01$ M, $E^\circ = 0.77$ V (given).

step 1 $\mathrm{Fe^{3+} + e^- \rightleftharpoons Fe^{2+}}$, $n = 1$.

step 2 $E = 0.77 - 0.0591 \log \frac{0.01}{0.1} = 0.77 + 0.0591.$

answer $\boxed{E = 0.8291 \text{ V}}$

check sympy: 0.829100 V. $n = 1$ doubles the correction compared with the tin couple.

added for clarity S003·7 · PDF p.67

W6.3 · Quinhydrone electrode at pH 4

ESA ★★

given $\mathrm{Pt|Q, QH_2}$ in a solution of pH 4 saturated with quinhydrone ($[Q] = [\mathrm{QH_2}]$); $E^\circ = 0.699$ V.

step 1 $\mathrm{Q + 2H^+ + 2e^- \rightleftharpoons QH_2}$, $n = 2$.

step 2 $E = 0.699 - \frac{0.0591}{2} \log \frac{[\mathrm{QH_2}]}{[\mathrm{Q}][\mathrm{H^+}]^2} = 0.699 - \frac{0.0591}{2} \log \frac{1}{(10^{-4})^2}.$

step 3 $= 0.699 - 0.02955 \times 8 = 0.699 - 0.0591 \times 4.$

answer $\boxed{E = 0.4626 \text{ V}}$

check sympy: 0.462600 V by both routes (E6.3 and E6.4).

S003·8 · T003 00:15:20

W6.4 · Model answer: notes on redox and gas electrodes (4 marks) ESA ★★

GIVEN ESA Dec 2023 Q2.d: Write a note on (i) oxidation–reduction electrodes (2M), (ii) gas electrode (2M).

STEP 1 (i) Inert Pt immersed in a solution containing the oxidised and reduced forms of a redox system; Pt is only an electrical contact; the potential arises from the tendency of one form to change into the other. Example $\mathrm{Pt|Sn^{4+}, Sn^{2+}}$: $\mathrm{Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}}$, $E = E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$.

STEP 2 (ii) A gas bubbled about an inert Pt foil immersed in a solution of the ions to which the gas is reversible; Pt gives contact and helps establish the gas–ion equilibrium. Example $\mathrm{Pt|H_2|H^+}$: $\mathrm{2H^+ + 2e^- \rightleftharpoons H_2}$, $E = E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[\mathrm{H^+}]^2}$; SHE $E = 0$.

ANSWER Definition + representation + reaction + Nernst equation for each (sketch G4.1 / G6.1 if time allows)

CHECK Every line is the slide wording of S003·5 and S003·7.

S003·5 · S003·7 · ESA Dec-23 2d

Past-paper questions

ESA Dec 2023 · 4 marks · UE22CY151B/A set 1, Q2.d · solved in W6.4

Write a note on the following types of electrodes: i. Oxidation-reduction electrodes (2M) ii. Gas electrode (2M).

SHORT-NOTE RECAP

Inert Pt in a solution of both forms of a redox couple ($\mathrm{Fe^{3+}/Fe^{2+}}$, $\mathrm{Ce^{4+}/Ce^{3+}}$, $\mathrm{Sn^{4+}/Sn^{2+}}$); Pt is only a contact; potential from the tendency of one form to change into the other: $E = E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$. Quinhydrone $\mathrm{Pt|Q, QH_2}$: $E^\circ = 0.699$ V, $E = 0.699 - 0.0591\,\mathrm{pH}$.

7 Ion-selective (membrane) electrode

An ion-selective electrode has a membrane in contact with the solution with which it can exchange ions, so it develops a potential that depends on the concentration of one particular ion in the presence of others. The glass electrode, used in pH meters, is selective to H^+ (also Na^+, K^+): $E_G = E_G^\circ + 0.0591 \log[H^+] = E_G^\circ - 0.0591\,\text{pH}$. ESA asks the definition and the cell with a calomel electrode; the pH numericals are Unit 3.

Key terms and definitions

Ion-selective electrode (membrane electrode). It consists of a membrane in contact with a solution, with which it can exchange ions; it develops a potential that depends on the concentration of one particular ion in the presence of other ions. E.g. the glass electrode, selective to H^+, Na^+, K^+. S003·9 · T003 00:16:05 · PDF p.44 · ESA Jun-24a 2b · ESA Jan-25 s1 2b

> *"Equation for determining potential for pH sensitive Glass electrode:* $E_G = E_G^0 + 0.0591 \log_{10}[H^+]$*."*
> S003·9

Formulas

E7.1 · Glass electrode (pH-sensitive ISE) ESA ★★

$$E_G = E_G^\circ + 0.0591 \log_{10}[H^+] = E_G^\circ - 0.0591\,\text{pH}$$

gives	Potential of the glass electrode.
applies	Glass membrane selective to H^+; 298 K.
units	V
note	E_G° differs for each electrode and is found by calibration with a buffer (Unit 3).

S003·9 · T003 00:17:47 · PDF p.56

E7.2 · Glass + calomel cell (PESU form) ESA ★★

$$E_{cell} = E_G - E_{cal} = E_G^\circ - 0.0591\,\text{pH} - E_{cal}$$

gives	The emf when a glass electrode is coupled with a calomel electrode.
applies	Calomel written on the left (anode), as in the consolidated PDF.
units	V
note	The pH numericals that use this are Unit 3 (class 36).

not on current slides PDF p.57 · PDF p.58 · ESA Jan-25 s1 2b

Graphs and diagrams

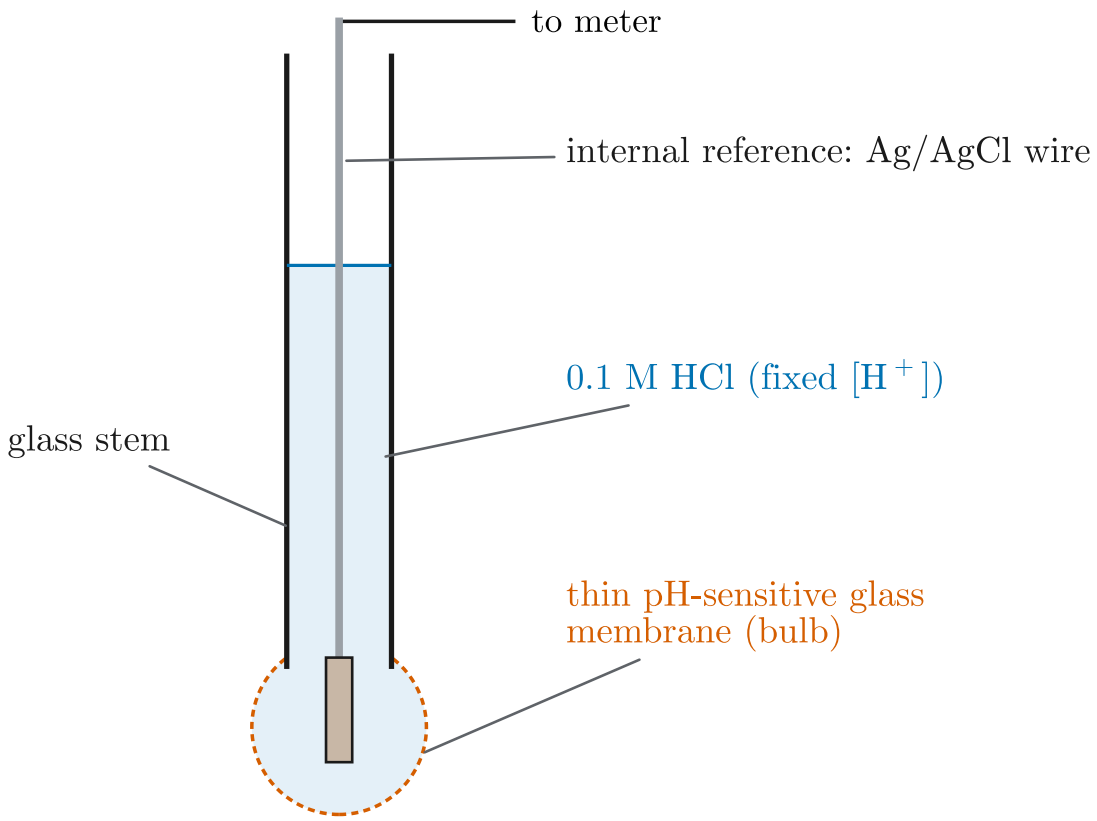


Figure G7.1. Glass electrode (simplified). ESA ★★

S003·9 · PDF p.56

How to read it. A glass tube ends in a thin pH-sensitive glass bulb (the membrane). Inside: 0.1 M HCl and an Ag/AgCl wire as the internal reference. The bulb dips in the test solution and is paired with an external reference such as the SCE. Construction details are taught in Unit 3.

Three electrodes whose potential depends on pH

	Hydrogen	Quinhydrone	Glass
Type	gas	redox (special case)	ion-selective
***E* at 298 K**	$-0.0591\,\mathrm{pH}$ (1 atm)	$0.699 - 0.0591\,\mathrm{pH}$	$E^\circ_G - 0.0591\,\mathrm{pH}$
Needs	H_2 gas, platinised Pt	quinhydrone, Pt	glass bulb + reference
Limit	poisoned easily; awkward gas supply	not above pH 8	calibrate E°_G with a buffer

ADDED FOR CLARITY S003·5 · S003·8 · S003·9

Important points and why

1. The membrane is the most important part: it exchanges ions with the solution and so develops a potential that depends on the concentration of one ion, even in the presence of other ions. ESA ★★ T003 00:16:43 · T003 00:18:14
2. Glass electrode: used in pH meters, selective to $\mathbf{H^+}$ (sometimes also $\mathbf{Na^+}$, $\mathbf{K^+}$). ESA ★★ S003·9 · T003 00:17:07
3. Other ISEs (older slides): crystalline $\mathbf{LaF_3}$ for $\mathbf{F^-}$, $\mathbf{Ag_2S}$ for $\mathbf{S^{2-}}$; liquid-membrane and PVC-immobilised ion exchangers for $\mathbf{Ca^{2+}}$, $\mathbf{K^+}$, $\mathbf{NO_3^-}$. ESA ★ NOT ON CURRENT SLIDES PDF p.45
4. A glass electrode cannot be used alone: it is coupled with an external reference such as the saturated calomel electrode, and the emf of that cell gives the pH (Unit 3). ESA ★★ NOT ON CURRENT SLIDES PDF p.57 · ESA Jun-24a 2b · ESA Jan-25 s1 2b

Why is it called ion-selective?
The membrane exchanges one particular ion, so the electrode responds to that ion's concentration and ignores (mostly) the others.
T003 00:16:05 · T003 00:16:25

Exam trap. Slide form $E_G = E_G^\circ + 0.0591 \log[\mathrm{H^+}]$ and pH form $E_G^\circ - 0.0591\,\mathrm{pH}$ are the same; mixing the sign of the pH term is the commonest error in the Unit 3 numericals. S003·9 · PDF p.56

Exam trap. In the glass + calomel cell, write E_{cell} to match your diagram: calomel on the left gives $E_{cell} = E_G - E_{cal}$ (PESU slides); glass on the left gives $E_{cell} = E_{cal} - E_G$. ADDED FOR CLARITY PDF p.57

Worked examples

W7.1 · ISE definition and the glass | SCE cell — ESA ★★

given ESA: What is an ion-selective electrode? Write the cell representation when a glass electrode is coupled with a saturated calomel electrode. (2 marks)

step 1 Definition: an electrode with a membrane in contact with the solution with which it can exchange ions; it responds selectively to one ion in the presence of others, e.g. the glass electrode (selective to H^+).

step 2 Cell (PESU form):
$Hg \,|\, Hg_2Cl_2(s) \,|\, KCl(sat) \,||\, H^+(\text{test solution}) \,|\, \text{glass} \,|\, 0.1\,M\,HCl \,|\, AgCl(s) \,|\, Ag.$

step 3 Emf: $E_{cell} = E_G - E_{SCE} = E^\circ_G - 0.0591\,\text{pH} - E_{SCE}$.

answer ISE as defined; cell as in step 2 (SCE anode, glass cathode)

check Matches the consolidated PDF p.57–58. The pH calculation that follows in both papers is Unit 3.

not on current slides · S003·9 · PDF p.57 · ESA Jun-24a 2b · ESA Jan-25 s1 2b

Past-paper questions

ESA Jun 2024 (a) · 2 marks · UE23CY151B/A set 1, Q2.b(i) · solved in W7.1

What is ion selective electrode? Write the cell representation when a glass electrode is coupled with a calomel electrode. (Part (ii), the pH numerical, is Unit 3.)

ESA Jan 2025 · 2 marks · UE24CY151A set 1, Q2.b(i) · solved in W7.1

What is ion selective electrode? Write the cell representation when a glass electrode is coupled with a saturated calomel electrode. (Part (ii), the pH numerical, is Unit 3.)

short-note recap

An ISE has a membrane that exchanges ions with the solution, so its potential depends on one ion's concentration in the presence of others. Glass electrode (pH meters): $E_G = E^\circ_G + 0.0591\log[H^+] = E^\circ_G - 0.0591\,\text{pH}$. With SCE: $Hg|Hg_2Cl_2|KCl(sat)||H^+|\text{glass}|0.1\,M\,HCl|AgCl|Ag$.

A Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
E	Electrode potential	Reduction potential of the electrode at the given concentrations	V	E1.1–E7.1
E°	Standard electrode potential	E with all activities 1 (1 M, 1 atm)	V	E1.1–E6.3
E_G, E°_G	Glass electrode potential	E°_G is a constant of each glass electrode	V	E7.1, E7.2
E_{cal}, E_{SCE}	Calomel potential	Saturated calomel 0.2422 V (PDF; papers use 0.2412–0.244 V)	V	E3.2, E7.2
n	Electrons transferred	From the balanced reduction reaction	—	E1.1–E6.3
R, T, F	Gas constant, temperature, Faraday	8.314 J/(K mol), 298 K, 96485 C/mol; $2.303RT/F = 0.0591$ V		E1.1, E2.1
[ ]	Concentration	Molar concentration (strictly activity)	mol/L	all
p_{H_2}, p_{Cl_2}	Partial pressure	Activity of a gas	atm	E4.1–E4.3
pH	pH	$-\log[H^+]$	—	E4.2, E6.4, E7.1
Q, QH_2	Quinone, hydroquinone	Oxidised and reduced forms in the quinhydrone electrode (not a reaction quotient)		E6.3, E6.4
Pb-Hg	Lead amalgam	Lead dissolved in mercury; activity below 1		E5.1
| and ||	Phase boundary, salt bridge	Single bar separates phases; double bar a salt bridge or junction		E7.2, W7.1

B Sources, coverage and gaps

Source	Path	Use
Deck 003 (.pptx rendered at 80 dpi; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.*	10 slides; content on 3–9
Transcript 003 (.transcript.md / .txt)	pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.*	18.6-minute lecture, read in full
Consolidated Unit 2 slides (older syllabus)	pesuc/First_Year_8enterprise/Files/Chemistry/Slides/Chem_Slides_Unit-2__1C1NjUHA4j.pdf	Cross-check of slides 17–24; ISE types (p.45); glass + calomel cell (p.56–58); calomel table (p.33)
ESA papers (13 files, 10 unique papers)	pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/*/*.pdf	3 in-scope questions (mapped in packs/_shared/chem/esa_map.md)
Video 003 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_*.mp4	Not opened: every slide is readable and the transcript covers it

Ten deduplicated ESA papers (Jul 2023 a/b, Dec 2023, Jan 2024 set 2, May 2024, Jun 2024 a/b, Dec 2024 UE23 and UE22, Jan 2025 sets 1 and 2) were checked. Class 003 is asked three times: Dec 2023 Q2.d (4 marks: notes on the oxidation–reduction electrode and the gas electrode), and Jun 2024a Q2.b(i) and Jan 2025 set 1 Q2.b(i) (define an ion-selective electrode and write the cell when a glass electrode is coupled with a (saturated) calomel electrode). The glass + SCE pH numerical that follows those questions was asked in 9 of the 10 papers, but in the 2026 syllabus it belongs to Unit 3 (pH sensor, class 36). The metal–metal ion, metal–insoluble salt, amalgam and quinhydrone electrodes were never asked on their own.

Known gaps, conflicts and corrections

1. No class-work (CW) sheet for Engineering Chemistry exists in the bucket, so the numericals are built from the deck's equations with standard data and checked with sympy; none comes with an official answer key.
2. The deck gives no standard potentials except $E^\circ_{Q/QH_2} = +0.699$ V. The values used in the worked examples (E° of Zn^{2+}/Zn −0.76 V, Ag/AgCl 0.222 V, Cl_2/Cl^- 1.36 V, Sn^{4+}/Sn^{2+} 0.15 V, Fe^{3+}/Fe^{2+} 0.77 V, lead amalgam −0.120 V) are stated as given data. 0.222 V and 0.77 V are the values used in the consolidated PDF (p.67).
3. pH numericals with the glass electrode and a calomel electrode (asked in 9 of the 10 papers) and the glass-electrode construction belong to Unit 3 (pH sensor, class 36) in the 2026 syllabus. This pack keeps only the ISE definition, the slide's equation $E_G = E^\circ_G + 0.0591 \log[H^+]$ and the cell representation that ESA asks here (W7.1).

4. The cell representation of glass + calomel is written in the consolidated PDF (p.57) with calomel on the left: $\mathrm{Hg|Hg_2Cl_2|Cl^-}$ || test solution |glass|0.1 N HCl|AgCl|Ag, with $E_{cell} = E_G - E_{cal}$. Many textbooks put the glass electrode on the left. Both are accepted if E_{cell} is written to match; W7.1 follows the PESU form.
5. Rendered slide 003·5: LibreOffice garbles the hydrogen-electrode Nernst equation (subscripts print as "PtTH"). It was read as $E = E^\circ - \frac{0.0591}{2} \log \frac{p_{H_2}}{[H^+]^2}$, which the lecture states word for word (T003 00:07:50).
6. Slide 003·7: the stannous–stannic reaction overlaps the line above it in the rendering; it is $\mathrm{Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}}$ (text export and T003 00:13:31).
7. Slide 003·8 draws quinone → hydroquinone with a one-way arrow; as an electrode it is an equilibrium (⇌), as the Nernst equation on the same slide assumes. The slide does not number the quinhydrone electrode; the lecture calls it a special case of the redox electrode (type 5).
8. Added for clarity (flagged): the pH forms $E = -0.0591\,\mathrm{pH}$ (hydrogen) and $E = 0.699 - 0.0591\,\mathrm{pH}$ (quinhydrone, $[Q] = [QH_2]$), the chlorine-electrode equation, the use of sodium amalgam (transcript only), the quinhydrone limit of pH 8, and the classification tree's three groups.
9. Deck 003 calls itself "Class 16 content" and the lecture says "module four": numbering from an older course plan; the bucket session number 003 is used here.
10. Transcript recognition errors were corrected silently: "Nant equation" → Nernst equation, "iron selective" → ion selective, "calomyl" → calomel, "Merkle" → mercury, "metaline electrode" → metal–metal ion electrode, "quin hydron" → quinhydrone, "hydroquinole" → hydroquinone, "silver plus iron" → silver ion, "sparingly soluble sort" → salt.

Coverage

Source	Citations	Slides cited (slide × citations)
S003	92	3×13, 4×11, 5×20, 6×8, 7×14, 8×13, 9×13
ESA	10	
PDF	15	
T003	66	

Slides 003·1, 003·2 and 003·10 are title, session and closing slides. The Nernst equations of slides 003·4–8 exist only as images; the text exports scatter them. The transcript covers every content slide (19 minutes).